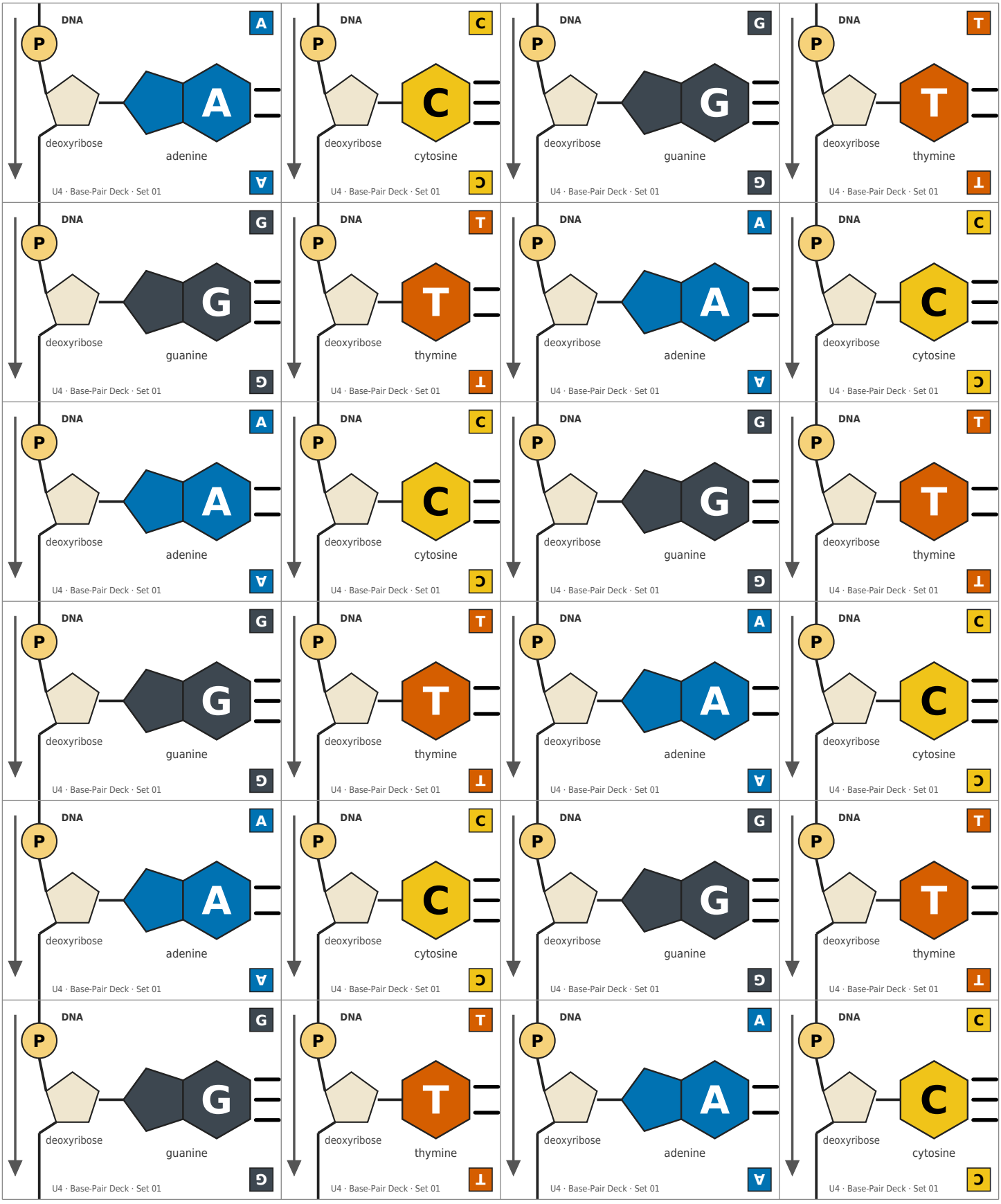
Unit 4 · Base-Pair Deck · Set 01

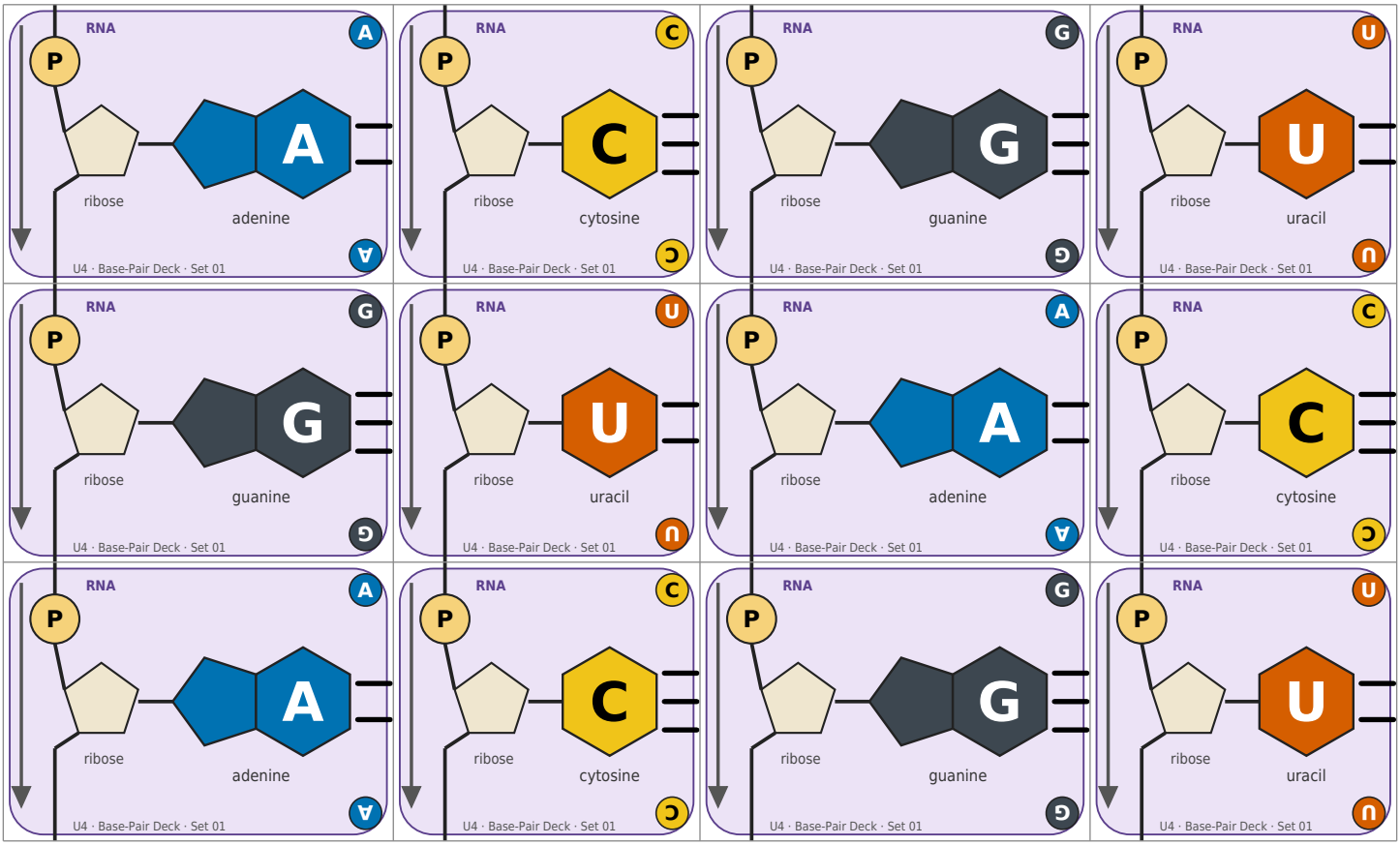
DNA cards (24) · card stock · print one-sided · cut on the grey lines



Canvas Original #68 · Unit 4 Base-Pair Deck · © J. R. Schwebach, CC BY-NC 4.0 · scienceonepagers.org

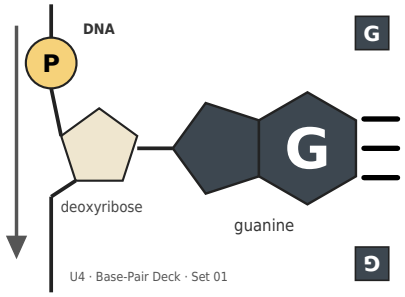
Unit 4 · Base-Pair Deck · Set 01

RNA cards (12) · tinted, round letter tiles · bag with the DNA cards of the same set

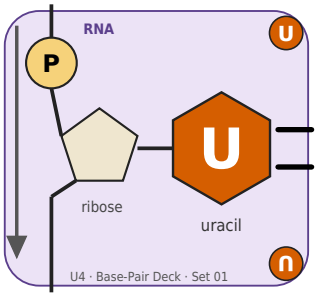


Unit 4 Base-Pair Deck · How to use it

Teacher card · one deck per pair of students · DNA structure, replication and transcription · Canvas Original #68



DNA card: white, square letter tiles, sugar labeled deoxyribose



RNA card: tinted, round letter tiles, sugar labeled ribose

What every card shows

- The three parts of a nucleotide: phosphate (circle, P), sugar (pentagon) and base (rings with the letter).
- The backbone line runs off the top and bottom edges, so stacked cards make one continuous sugar–phosphate rail.
- Ring count. Adenine and guanine (purines) have two rings and longer cards. Cytosine, thymine and uracil (pyrimidines) have one ring and shorter cards. Only a two-ring card facing a one-ring card makes a rung the same width as every other rung.
- Bond marks on the facing edge: two on A, T and U, three on G and C. In a correct pair the marks meet line for line.
- Arrow on the backbone edge. A partner card is laid turned all the way around (180°), so its arrow runs the other way: the two strands are antiparallel. The corner letter tiles read upright either way.

What is in one set (36 cards, one zip bag)

- DNA: 24 cards, 6 each of A, T, C and G. RNA: 12 cards, 3 each of A, U, C and G.
- Every card carries the line U4 · Base-Pair Deck · Set NN, so a stray card goes back to its own bag. Label each bag with its set number and count cards at the end of class.

Printing and cutting

- Letter card stock, one-sided, nothing on the back. Each set is one DNA sheet plus half an RNA sheet; 18 sets = 18 DNA sheets + 9 RNA sheets = 27 sheets. Color is best; in black and white the letters, ring count and bond marks still carry everything.
- A paper cutter is fastest: the grey lines run straight across the whole sheet (inside the border: 3 cuts down and 5 across on a DNA sheet). Cut each RNA sheet in half on the dashed line first.

Lesson 1 · DNA structure (LT 4.1): build a ladder

- Pairs lay six DNA cards in a column, arrows all pointing the same way, then build the second strand beside it.
- Prompts: “Which cards make a rung the same width as the others?” “Count the bond marks on each side.” “Which way does your second strand’s arrow point?” Let students find the pairing rule from the fit before you name it.
- Name it afterward: a two-ring base always pairs with a one-ring base, which keeps the ladder the same width all the way up.

Lesson 2 · Replication (LT 4.2): unzip and copy

- Build a six-rung double strand, then slide the two strands apart at the bond marks.
- Each partner takes one old strand and builds a new partner for it from the spare cards. Compare the two finished molecules with each other and with the original.
- Ask: “What does each new molecule keep from the original?” (one old strand and one new strand).

Lesson 4 · Transcription (LT 4.3): why the mRNA matches the coding strand

- Build a nine-rung double strand whose top strand begins with the start sequence. Unzip it and name the template strand.
- Lay RNA cards along the template, pairing base by base; an A on the template takes a U.
- Lift the RNA row and set it over the top (coding) strand. Ask: “What do you notice?” Land it with the photo negative: the template is the negative, so the print made from it looks like the original.

Advanced extra

- The arrow runs from the phosphate end to the sugar end of each nucleotide, which is the 5′ → 3′ direction. Advanced students can label each strand 5′ at the phosphate end and 3′ at the other, and see that the two strands run opposite ways.

If the cards are not ready

- Run Lesson 1 from the packet’s ladder diagram and start the cards in Lesson 2.

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